Problem 5

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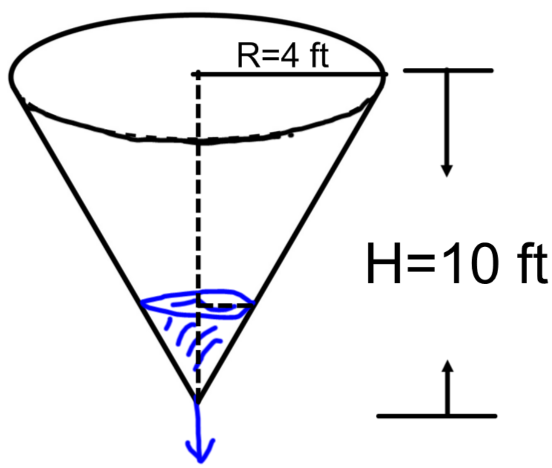
Problem 1 *A conical water tank of height 10 feet and radius 4 feet is being drained through a hole in the bottom (See figure below).*



(a) *Let r be the radius of the water at depth h. In the figure above, label r and h.*

(b) *Find an expression for r in terms of h.*

(c) *The water in the tank is being drained at a rate of 2 feet^3 per minute. What is the rate of change of the water depth when the water depth is 3 feet? [3]*